A two-dimensional counterexample to p-orthonormalization in Banach spaces

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2026-06-14

Source problem

The source paper is K. Mahesh Krishna, *Feichtinger Conjectures, R_ε-Conjectures and Weaver's Conjectures for Banach spaces*, arXiv:2201.00125. Problem 2.6 appears on page 5 of the source PDF.

> Gram-Schmidt orthonormalization converts every linearly independent set in a Hilbert space into an orthonormal set and hence to an orthonormal basis [55,88]. On the other hand, there is a Gram-Schmidt orthonormalization in Banach spaces due to Lin [92]. We ask the following open problem.
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> **Problem 2.6.** ***Let $\{\tau_j\}_{j=1}^n$ be a linearly independent collection in $\mathcal{X}$ and $\{f_j\}_{j=1}^n$ be a linearly independent collection in $\mathcal{X}^*$. Whether there is a way to convert $(\{f_j\}_{j=1}^n, \{\tau_j\}_{j=1}^n)$ into a p-orthonormal sequence, say $(\{g_j\}_{j=1}^n, \{\omega_j\}_{j=1}^n)$ such that***
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> $$\operatorname{span}\{f_1, \dots, f_j\} = \operatorname{span}\{g_1, \dots, g_j\} \quad \textbf{\textit{and/or}} \quad \operatorname{span}\{\tau_1, \dots, \tau_j\} = \operatorname{span}\{\omega_1, \dots, \omega_j\}, \quad \forall 1 \le j \le n.$$
>
> ***In particular, whether we can convert a $(\{f_j\}_{j=1}^n, \{\tau_j\}_{j=1}^n)$ into a p-orthonormal basis?***

In words, Problem 2.6 asks whether, for linearly independent collections $\{\tau_j\}_{j=1}^n \subset X$ and $\{f_j\}_{j=1}^n \subset X^*$, one can convert the pair into a p-orthonormal sequence $(\{g_j\}_{j=1}^n, \{\omega_j\}_{j=1}^n)$ preserving the initial spans of the functionals and/or vectors. It also asks whether such data can be converted into a p-orthonormal basis.

Relevant definition

For a subset $M \subseteq \mathbb{N}$, a pair $(\{g_n\}_{n\in M}, \{\omega_n\}_{n\in M})$ is a p-orthonormal sequence in X in the sense of the source paper if:

1. $g_n(\omega_m) = \delta_{n,m}$ for all $n, m \in M$;
2. $\|x\|^p \ge \sum_{n\in M} |g_n(x)|^p$ for every $x \in X$;
3. $\left\|\sum_{n\in M} a_n\omega_n\right\|^p = \sum_{n\in M} |a_n|^p$ for every $(a_n)_{n\in M} \in \ell^p(M)$.

Proof Intuition

Condition (3) says that the selected vectors ω_n span an isometric copy of $\ell_p^{|M|}$. For $p = 1$ and two vectors this is impossible inside the Euclidean plane: two unit Euclidean vectors whose sum has norm 2 must point in exactly the same direction. But condition (1) requires them to be biorthogonal, so they cannot coincide.

Counterexample. *Problem 2.6 has a negative answer as stated. In the real Euclidean plane $X = (\mathbb{R}^2, \|\cdot\|_2)$ with $p = 1$, there is no two-term 1-orthonormal sequence. Consequently, no procedure can convert arbitrary two-term independent data into a 1-orthonormal sequence preserving either family of spans.*

Proof. Let $X = (\mathbb{R}^2, \|\cdot\|_2)$ and $p = 1$. Take, for instance,

$$\tau_1 = (1, 0), \qquad \tau_2 = (0, 1),$$

and let $f_1(x, y) = x$, $f_2(x, y) = y$. These are linearly independent collections in X and X^*.

Suppose that a two-term 1-orthonormal sequence $(g_1, g_2; \omega_1, \omega_2)$ existed in X. Applying condition (3) with $(a_1, a_2) = (1, 0)$ and $(0, 1)$ gives

$$\|\omega_1\|_2 = \|\omega_2\|_2 = 1.$$

Applying condition (3) with $(a_1, a_2) = (1, 1)$ gives

$$\|\omega_1 + \omega_2\|_2 = 2.$$

Equality in the Euclidean triangle inequality for two unit vectors implies $\omega_1 = \omega_2$. But condition (1) gives

$$g_1(\omega_1) = 1, \qquad g_1(\omega_2) = 0,$$

which is impossible if $\omega_1 = \omega_2$. Hence no two-term 1-orthonormal sequence exists in the Euclidean plane.

Therefore the independent data $(f_1, f_2; \tau_1, \tau_2)$ cannot be converted into a 1-orthonormal sequence of length two, regardless of whether one asks to preserve the functional flags, the vector flags, or both. The stronger "in particular" request for conversion into a p-orthonormal basis also fails for this example. $\square$

Verification notes

The example satisfies the hypotheses of Problem 2.6 with $n = 2$: the two vectors τ_1, τ_2 are linearly independent, and the two coordinate functionals f_1, f_2 are linearly independent. The contradiction uses only the source paper's own definition of a p-orthonormal sequence and the standard equality case of the Euclidean triangle inequality.

No computational check is needed. A bounded novelty check was performed on 2026-06-14: the run indexes were searched for arXiv:2201.00125 and core terms "Feichtinger", "Weaver", and "p-orthonormal"; the source paper was searched around Problem 2.6 and the relevant definitions; web queries for "p-orthonormal sequence Feichtinger Banach counterexample", "2201.00125 p-orthonormal", and the exact phrase "Whether there is a way to convert" with "p-orthonormal sequence" returned no direct answer. Novelty confidence is therefore moderate rather than high, since the obstruction is elementary.

Limitations

This packet disproves Problem 2.6 as written for arbitrary Banach spaces and arbitrary $p \in [1, \infty)$. It does not address possible restricted versions, for example spaces containing isometric copies of ℓ_p^n, or the Hilbert case $p = 2$ where the usual Gram-Schmidt mechanism is available.

Human-review recommendation

Check that Problem 2.6 is intended to include the choice $p = 1$ and the Euclidean plane. Under that literal reading, the counterexample is complete.

References

[1] K. Mahesh Krishna, *Feichtinger Conjectures, R_ε-Conjectures and Weaver's Conjectures for Banach spaces*, arXiv:2201.00125, 2022.